Lecture 05 – Variational Bayesian

Stat 5376: Statistical Machine Learning

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Frequentist Framework

- Maximize likelihood: $\max_\theta p(Z_{1:n}|\theta) = \prod_{i=1}^n p(Z_i|\theta)$
  - Equivalent to $\min_\theta L(\theta) = -\frac{1}{n}\log p(Z_{1:n}|\theta) = -\frac{1}{n}\sum_{i=1}^n \log p(Z_i|\theta)$
  - The maximizer/minimizer is the frequentist estimator.
- Maximal likelihood estimation (MLE):
  - $Z \sim Bin(1,\theta)$: $P(Z=1)=\theta, P(Z=0)=1-\theta$
  - $p(z|\theta) = \theta^z(1-\theta)^{1-z}$
  - Log likelihood:

$$\sum_{i=1}^n \log p(Z_i|\theta) = \sum_{i=1}^n Z_i \log(\theta) + \left(n - \sum_{i=1}^n Z_i\right)\log(1-\theta)$$

  - Equivalent to

$$\min_\theta L(\theta) = -\overline{Z}\log(\theta) - \left(1-\overline{Z}\right)\log(1-\theta),$$

$$L'(\theta) = -\frac{\overline{Z}}{\theta} + \frac{1-\overline{Z}}{1-\theta} = 0$$

  - Frequentist estimator: $\widehat{\theta} = \overline{Z}$

Frequentist Framework

- Regression:
  - $Z = (Y, X)$
  - $Y|X \sim N(f_\theta(X), \sigma^2)$ and $X \sim g(x)$
  - $p(z|\theta) = p(y, x|\theta) = p(y|x, \theta)g(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y - f_\theta(x))^2}{2\sigma^2}} \times g(x)$
  - Log likelihood:

$$\sum_{i=1}^{n} \log p(Y_i, X_i|\theta) = -\frac{1}{2\sigma^2} \sum_{i=1}^{n} (Y_i - f_\theta(X_i))^2 - n \log(\sqrt{2\pi}\sigma) + \sum_{i=1}^{n} \log g(X_i)$$

  - Equivalent to

$$\min_\theta L(\theta) = \frac{1}{n} \sum_{i=1}^{n} (Y_i - f_\theta(X_i))^2$$

  - Frequentist estimator: no analytic expression, use pytorch to solve the problem

Bayesian Framework

- Bayesian procedure:
  - Likelihood: $p(Z_{1:n}|\theta) = \prod_{i=1}^{n} p(Z_i|\theta)$
  - Prior: $\theta \sim \pi(\theta)$
  - Joint distribution: $p(Z_{1:n}, \theta) = p(Z_{1:n}|\theta)\pi(\theta)$
  - Posterior:

$$p(\theta|Z_{1:n}) = \frac{p(Z_{1:n}, \theta)}{\int p(Z_{1:n}, \theta)d\theta} = \frac{p(Z_{1:n}|\theta)\pi(\theta)}{\int p(Z_{1:n}|\theta)\pi(\theta)d\theta} = \frac{p(Z_{1:n}|\theta)\pi(\theta)}{m(Z_{1:n})}$$

  - Bayesian estimator:

$$\widehat{\theta}_{\mathrm{BE}} = \int \theta p(\theta|Z_{1:n})d\theta,$$

    which is the posterior mean.
- The purpose of Bayesian is to calculate the posterior $p(\theta|Z_{1:n})$.

Bayesian Framework

- Maximal likelihood estimation (MLE):
  - $Z \sim Bin(1, \theta)$: $P(Z = 1) = \theta, P(Z = 0) = 1 - \theta$
  - $p(z|\theta) = \theta^z (1 - \theta)^{1-z}$
  - Likelihood: $p(Z_{1:n}|\theta) = \prod_{i=1}^n \theta^{Z_i}(1 - \theta)^{1-Z_i} = \theta^{n\overline{Z}}(1 - \theta)^{n-n\overline{Z}}$
  - Prior: $\theta \sim \pi(\theta) = Unif(0, 1)$
  - Joint distribution: $p(Z_{1:n}, \theta) = p(Z_{1:n}|\theta)\pi(\theta) = p(Z_{1:n}|\theta)$
  - Posterior:

$$m(Z_{1:n}) = \int_0^1 p(Z_{1:n}|\theta)d\theta = B(n\overline{Z} + 1, n - n\overline{Z} + 1),$$

$$p(\theta|Z_{1:n}) = \frac{1}{B(n\overline{Z} + 1, n - n\overline{Z} + 1)}\theta^{n\overline{Z}}(1 - \theta)^{n-n\overline{Z}}.$$

  - Posterior is $Beta(n\overline{Z} + 1, n - n\overline{Z} + 1)$
  - Bayesian estimator: $\widehat{\theta}_{\mathrm{BE}} = \frac{n\overline{Z}+1}{n+2}$
    - Recall the mean of $Beta(\alpha, \beta)$ is $\frac{\alpha}{\alpha+\beta}$
  - We are lucky since we use uniform prior, so the posterior is easy.
  - What if π is arbitrary?

Bayesian Framework

- Regression:
  - $Z = (Y, X)$
  - $Y|X \sim N(f_\theta(X), \sigma^2)$ and $X \sim g(x)$
  - $p(z|\theta) = p(y, x|\theta) = p(y|x, \theta)g(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y - f_\theta(x))^2}{2\sigma^2}} \times g(x)$
  - Likelihood:

$$p(Z_{1:n}|\theta) = \frac{1}{(\sqrt{2\pi}\sigma)^n} e^{-\sum_{i=1}^n \frac{(Y - f_\theta(X_i))^2}{2\sigma^2}} \times \prod_{i=1}^n g(X_i)$$

  - Prior: $\theta \sim \pi(\theta)$
  - Posterior: $p(\theta|Z_{1:n}) = \frac{p(Z_{1:n}|\theta)\pi(\theta)}{\int p(Z_{1:n}|\theta)\pi(\theta)d\theta}$, no analytic expression

Bayesian Framework

- Bayesian procedure:
  - Likelihood: $p(Z_{1:n}|\theta) = \prod_{i=1}^{n} p(Z_i|\theta)$
  - Prior: $\theta \sim \pi(\theta)$
  - Joint distribution: $p(Z_{1:n}, \theta) = p(Z_{1:n}|\theta)\pi(\theta)$
  - Posterior:

$$p(\theta|Z_{1:n}) = \frac{p(Z_{1:n}, \theta)}{\int p(Z_{1:n}, \theta)d\theta} = \frac{p(Z_{1:n}|\theta)\pi(\theta)}{\int p(Z_{1:n}|\theta)\pi(\theta)d\theta} = \frac{p(Z_{1:n}|\theta)\pi(\theta)}{m(Z_{1:n})}$$

- $p(\theta|Z_{1:n})$ is difficult to calculate because of $m(Z_{1:n})$.
  - Markov chain Monte Carlo (MCMC):
    - sampling from the true posterior $p(\theta|Z_{1:n})$
    - Slow if θ is high-dimensional or sample size n is large
    - Unbiased
  - Variational Bayesian:
    - Try to approximate it by $q(\theta)$
    - Example: $q(\theta)$ is normal
    - Biased if the distribution class is not carefully chosen
    - Faster and scales well with big data (use SGD)

Variational Bayesian

- Similarity between distributions:
  - Kullback-Leibler (KL) divergence:

$$KL(q,p) = \int q(\theta) \log\left(\frac{q(\theta)}{p(\theta)}\right) d\theta$$

  - KL divergence is measure the dissimilarity between two distributions q and p
  - Note: $KL(q,p) \neq KL(p,q)$, $KL(q,p) \geq 0$ and $KL(p,p) = 0$.
- $p(\theta|Z_{1:n})$ is difficult to calculate
- Try to approximate it by $q(\theta)$ by

$$\min_{q \in Q} KL(q(\theta), p(\theta|Z_{1:n})),$$

  where Q is a class of distributions.
- Gaussian family: $Q = \{q_{\mu,\kappa}(\theta) : q_{\mu,\kappa} = N(\mu, \kappa^2), \mu \in \mathbb{R}, \kappa > 0\}$

Variational Bayesian

- Recall $p(\theta|Z_{1:n}) = \frac{p(Z_{1:n}|\theta)\pi(\theta)}{m(Z_{1:n})}$, we have

$$
\begin{aligned}
KL(q(\theta), p(\theta|Z_{1:n})) &= \int q(\theta) \log\left(\frac{q(\theta)}{p(\theta|Z_{1:n})}\right) d\theta \\
&= \int q(\theta) \log\left(\frac{q(\theta)m(Z_{1:n})}{p(Z_{1:n}|\theta)\pi(\theta)}\right) d\theta \\
&= \int q(\theta) \log\left(m(Z_{1:n})\right) d\theta + \int q(\theta) \log\left(\frac{q(\theta)}{\pi(\theta)}\right) d\theta \\
&\quad - \int q(\theta) \log\left(p(Z_{1:n}|\theta)\right) d\theta \\
&= m(Z_{1:n}) + KL(q, \pi) - \mathbb{E}_q\left(\log(p(Z_{1:n}|\theta))\right).
\end{aligned}
$$

- Minimizing $KL(q(\theta), p(\theta|Z_{1:n}))$ is equivalent to maximizing

$$
ELBO(q) = \mathbb{E}_q\left(\log(p(Z_{1:n}|\theta))\right) - KL(q, \pi)
$$

- ELBO: Evidence lower bound
- $\mathbb{E}_q(g(\theta)) = \int g(\theta)q(\theta)d\theta$, the expectation of $g(\theta)$ under $\theta \sim q(\theta)$

Variational Bayesian – Computation

- $ELBO(q) = \mathbb{E}_q\left(\log(p(Z_{1:n}|\theta))\right) - KL(q,\pi)$
- Monte Carol:
  - $\mathbb{E}_q(g(\theta)) = \int g(\theta)q(\theta)d\theta \approx \frac{1}{M}\sum_{i=1}^{M} g(\theta_i)$ for $\{\theta_i\}_{i=1}^{M}$ drawn from $q(\theta)$
- First term:
  - Draw $\theta_1, \ldots, \theta_M$ from $q(\theta)$
  - $\mathbb{E}_q\left(\log(p(Z_{1:n}|\theta))\right) \approx \frac{1}{M}\sum_{i=1}^{M} \log p(Z_{1:n}|\theta_i)$
- Second term:
  - If $KL(q,\pi)$ is easy to calculate, just calculate it by hand.
  - Example: $q \sim N(\mu_q, \sigma_q^2), \pi \sim (\mu_\pi, \sigma_\pi^2)$, then

$$KL(q,\pi) = \log\frac{\sigma_\pi}{\sigma_q} + \frac{\sigma_q^2 + (\mu_q - \mu_\pi)^2}{2\sigma_\pi^2} - \frac{1}{2}.$$

  - If it is difficult to calculate, use Monte Carol:

$$\begin{aligned} KL(q,\pi) &= \int q(\theta)\log\left(\frac{q(\theta)}{\pi(\theta)}\right)d\theta \\ &= \mathbb{E}_q\left(\log\left(\frac{q(\theta)}{\pi(\theta)}\right)\right) \approx \frac{1}{M}\sum_{i=1}^{M}\log\left(\frac{q(\theta_i)}{\pi(\theta_i)}\right). \end{aligned}$$

Variational Bayesian – Computation

- Minimize the KL divergence over $Q = \{q_\eta : \eta \in \mathbb{R}^k\}$
  - η is called variational parameter
  - It reduces the optimization problem from finding a function to a vector
- Example: $Q = \{q_{\mu,\kappa}(\theta) : q_{\mu,\kappa} = N(\mu, \kappa^2), \mu \in \mathbb{R}, \kappa > 0\}$ with $\eta = (\mu, \sigma)$
- $ELBO(q_\eta) = \mathbb{E}_{q_\eta}\left(\log(p(Z_{1:n}|\theta))\right) - KL(q_\eta, \pi)$
- Optimization:
  (i) Initialize η
  (ii) Generate $\theta_1^{(\eta)}, \ldots, \theta_M^{(\eta)}$ from q_η
  (iii) Use one the following equation

$$ELBO(q_\eta) = \begin{cases} \frac{1}{M}\sum_{i=1}^M \log p(Z_{1:n}|\theta_i^{(\eta)}) - KL(q_\eta, \pi), \\ \frac{1}{M}\sum_{i=1}^M \log p(Z_{1:n}|\theta_i^{(\eta)}) - \frac{1}{M}\sum_{i=1}^M \log\left(\frac{q_\eta(\theta_i^{(\eta)})}{\pi(\theta_i^{(\eta)})}\right). \end{cases}$$

  (iv) Gradient method to update η based on $ELBO(q_\eta)$
  (v) Repeat (ii)-(iv) till convergence

Variational Bayesian – Computation

- Let $\widehat{\eta}$ be the maximizer of $ELBO$, then $q_{\widehat{\eta}}$ is the approximated posterior
- Bayesian estimator:

$$\widehat{\theta}_{\mathrm{BE}} = \int \theta q_{\widehat{\eta}}(\theta) d\theta \approx \frac{1}{M} \sum_{i=1}^{M} \theta_i,$$

  where $\theta_1, \ldots, \theta_M$ from $q_{\widehat{\eta}}$.
- Credible interval/band: we also can use the variance/sd of $\theta_1, \ldots, \theta_M$ to quantify the uncertainty of the parameter.
- Credible interval is not confidence interval

Temperature Parameter in Regression

- Frequentist loss: $\min_\theta L(\theta) = \frac{1}{n}\sum_{i=1}^n (Y_i - f_\theta(X_i))^2$
- Variational Bayesian:
  - Likelihood:

$$\sum_{i=1}^n \log p(Y_i, X_i|\theta) = -\frac{1}{2\sigma^2}\sum_{i=1}^n (Y_i - f_\theta(X_i))^2 - n\log(\sqrt{2\pi}\sigma) + \sum_{i=1}^n \log g(X_i)$$

  - ELBO:

$$\begin{aligned}\max_{q\in Q} ELBO(q) &= \mathbb{E}_q\left(\log(p(Z_{1:n}|\theta))\right) - KL(q,\pi)\\ &= -\frac{1}{2\sigma^2}\sum_{i=1}^n \mathbb{E}_q\left((Y_i - f_\theta(X_i))^2\right) - KL(q,\pi),\end{aligned}$$

  - $\sigma > 0$: temperature parameter
  - Need to specify σ.
  - Temperature parameter behaves like penalty parameter (λ in lasso):

$$\min_{q\in Q}\sum_{i=1}^n \mathbb{E}_q\left((Y_i - f_\theta(X_i))^2\right) + \underbrace{2\sigma^2 KL(q,\pi)}_{\text{penalty term}}$$

Variational Bayesian – Mean-Field Approximation

- Naive idea:

$$q_\eta(\theta_1, \theta_2) = N\left(\begin{pmatrix}\mu_1\\ \mu_2\end{pmatrix}, \begin{pmatrix}\kappa_1^2, \kappa_{12}\\ \kappa_{12}, \kappa_2^2\end{pmatrix}\right).$$

  There are 5 parameters.

- Mean-field approximation:

$$q_\eta(\theta_1, \theta_2) = N\left(\begin{pmatrix}\mu_1\\ \mu_2\end{pmatrix}, \begin{pmatrix}\kappa_1^2, 0\\ 0, \kappa_2^2\end{pmatrix}\right)$$

  There are 4 parameters.

- K-dimension: $2 + (K^2 + K)/2$ V.S. $2K$ parameters

Variational Bayesian – Mean-Field Approximation

- One-dimension: $Q = \{q_{\mu,\kappa}(\theta) : q_{\mu,\kappa} = N(\mu, \kappa^2), \mu \in \mathbb{R}, \kappa > 0\}$ with $\eta = (\mu, \kappa)$
- Two-dimension:

$$Q = \left\{q_{\mu_1,\kappa_1}(\theta_1) q_{\mu_2,\kappa_2}(\theta_2) : q_{\mu_i,\kappa_i} = N(\mu_i, \kappa_i^2), \mu_i \in \mathbb{R}, \kappa_i > 0\right\},$$

  with $\eta = (\mu_1, \kappa_1, \mu_2, \kappa_2)$
- Multiple-dimension:

$$Q = \left\{\prod_{i=1}^{K} q_{\mu_i,\kappa_i}(\theta_i) : q_{\mu_i,\kappa_i} = N(\mu_i, \kappa_i^2), \mu_i \in \mathbb{R}, \kappa_i > 0\right\},$$

  with $\eta = (\mu_1, \kappa_1, \ldots, \mu_K, \kappa_K)$

Variational Bayesian – Reparameterization Trick

- The random draw is not differentiable:

```
z = torch.normal(mu, kappa)   # sample from N(mu, kappa^2)
loss = f(z)
loss.backward()
```

- Reparameterization:

```
mu = torch.tensor(0.0, requires_grad=True)
logvar = torch.tensor(0.0, requires_grad=True)   # log(kappa^2)
kappa = torch.exp(0.5 * logvar)
eps = torch.randn()    # sample from N(0, 1)
z = mu + kappa * eps # Reparameterize to N(mu, kappa^2)
loss = f(z)
loss.backward()
```

- Two tricks:
  - $z = N(0, 1) \times \kappa + \mu$, so that we can find the derivatives of κ and μ
  - Use the parameter $\mathrm{logvar} = \log(\kappa^2)$ instead of κ to avoid constrain $\kappa > 0$.

Variational Bayesian – Coding

- See instruction in class

More about Variational Bayesian

- When θ is continuous, we use normal variational family to approximate the posterior.
  - When θ is discrete, such as categorical weight prior $\pi(\theta_i = k) = w_k$ for $k = 1, \ldots, m$, we use categorical variational family $q(\theta_i = k) = \eta_k$ for $k = 1, \ldots, m$.
  - The variational parameter is $\eta = (\eta_1, \ldots, \eta_m)$.
  - Reparameterization trick is more complicate, Gumbel–Softmax trick.
- A convenient python package: pyro
- An important application: Variational Autoencoder (VAE)